

AI Project 3: AI Use in Teaching Transport

Due: 10/14/25

Overview

Artificial Intelligence (AI) is increasingly useful not only for solving transport problems, but also as a tool for studying and consolidating concepts. This project is designed to help you prepare for the upcoming midterm (10/16) while developing skills in AI-assisted problem solving, technical writing in L^AT_EX, and oral presentation.

Assignment

You are asked to use AI to study and analyze the Spring 2021 midterm (posted on the class website). For each problem on that exam:

1. **AI Solution in L^AT_EX.** Use AI to prepare a L^AT_EX-formatted solution to the problem. You may edit the AI output as needed for clarity, correctness, and formatting.
2. **Concept Analysis.** Write a short discussion identifying the main concepts being tested in the problem. Be specific (e.g., “unsteady diffusion with similarity variables” rather than “diffusion”).
3. **Study Guide.** Provide a short study guide for the concepts behind the problem. This should include:
 - 2–3 key concepts students must master.
 - References to BSL (sections/examples), class notes, or reputable web sources with verified url.
 - At least one additional practice problem or variation (state the problem, do not solve it).
4. **AI Interaction Notes (optional but encouraged).** Include in an appendix selected excerpts of your AI prompts and responses, especially where you had to refine or correct the output. Briefly explain any corrections or adjustments you made.
5. **Screen Recording.** Create a narrated screen recording (similar to recorded lectures). In your recording:
 - Explain the solution to each problem.
 - Highlight the conceptual analysis and study guide elements.
 - Emphasize your reasoning, not just reading the AI-generated text.

Upload the recording to Panopto and insert the link(s) into both your L^AT_EX source and PDF.

Deliverables

- A compiled PDF report containing:
 - a. A restatement of each problem.
 - b. Solutions in $\text{\LaTeX}$ for each problem.
 - c. Concept analysis and study guide for each problem.
 - d. Panopto link(s) to your screen recording.
 - e. Optional appendix with AI interaction notes.
- Your Panopto recording uploaded and linked into the PDF.

Guidelines

- Keep your total recording time under 30 minutes. Aim for about 5–7 minutes per problem.
- Remember: the midterm will be closed-book and closed-AI. The purpose of this project is to use AI as a learning partner now, so you are prepared to work independently on the exam.
- Professionalism in both your $\text{\LaTeX}$ write-up and recorded presentation will be part of the evaluation.

Rubric

Your grade will be based on the following criteria:

1. **Accuracy of Solutions (30%)** Correctness of the $\text{\LaTeX}$ -formatted solutions and appropriateness of any edits to AI output.
2. **Depth of Conceptual Analysis (25%)** Clarity in identifying the key transport phenomena concepts each problem is testing.
3. **Usefulness of Study Guide (25%)** Quality of references, clarity of explanations, and inclusion of at least one meaningful practice problem per question.
4. **Clarity and Professionalism of $\text{\LaTeX}$ Write-Up (10%)** Organization, readability, formatting, and integration of links.
5. **Quality of Recorded Presentation (10%)** Effectiveness of narration, clarity of reasoning, and adherence to the time guideline.